

Introduction to Text Processing

Social and Cultural Analytics (7AAVBCS5)

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Lecture Plan

- **Introduction**

- Quick review — the analytical pipeline
- The summative

- **Part I: Introduction to NLP**

- **Part II: The NLP Pipeline**

- Pre-processing, tokenisation, lemmatisation, information extraction

- **Part III: spaCy**

- **Part IV: Live Coding**

Key Learning Outcomes

By the end of this week you will be able to:

- Understand what NLP is and why it matters for cultural analytics
- Describe the typical NLP processing pipeline
- Pre-process text (lower-casing, stop-word removal)

- Tokenise text, and understand types vs tokens
- Create and analyse word frequency lists in Python
- Apply text processing to research questions

The analysis process

Stage	What you are doing	Tools
Inspect	what is the data, what does it contain and represent?	<code>read_csv</code> , <code>type</code> , <code>head</code> , <code>shape</code> , <code>info</code>
Summarise	min, mean, median, max, IQR	<code>describe</code>
Process	fix what needs fixing	<code>dropna</code> , <code>fillna</code> , <code>astype("category")</code>
Descriptive	the shape of the data	<code>value_counts</code> , <code>crosstab</code> , <code>groupby</code>
Analysis	test ideas, look for relationships	<code>corr</code> , <code>scatter</code> , <code>bar</code> , <code>hist</code>

You may need to go back to processing. Nobody does this in one pass.

The Summative

Purpose and scope

Purpose

- You are designing and carrying out a data analysis project in a domain of your choice
- It must sit within **digital culture and society** — not generic data science, finance or marketing
- Not a traditional essay
- You are writing for an informed community of analysts, not an academic journal

Scope and flexibility

- You can change your project from the **data overview** — that was formative
 - Last year's examples are useful only for structure and technical style
- Not a technical tutorial — but your code must be

understandable and justified.

Format

- A short report: **3,500 words** ($\pm 15\%$), **code included in the word count**
- Due **21 January 2027**
- Must be generated automatically from a **notebook** — no screenshots of code
- Include all code in the notebook
- The only editing allowed after export is adding the cover sheet

Structure

Introduction — objectives and research questions; 1–2 paragraphs of background; 5–10 references; situate in digital culture/society

Data — what datasets, why they fit your questions, how you accessed them

Analysis — the steps, aligned with your research questions. Code correct and **interpreted in the text**. Fine if not all questions are answerable — justify why

Results & interpretation — findings in relation to your questions; who benefits, and what the analysis reveals about digital culture/society; **include limitations**

Conclusion — what you did, how you analysed it,

what the results show.

Future work optional

Assessment criteria

You must demonstrate that:

1. The use case is clearly described and connected to digital culture/society
2. Research questions/objectives are explicit
3. Datasets are appropriate, clearly described, and properly accessed
4. Analysis is relevant to the questions and clearly presented
5. Code is correct, integrated, and explained in the text
6. Interpretation is strong — what do the results mean?
7. Impact is articulated — who benefits and why?
8. Limitations are acknowledged where relevant

Points 3, 6 and 8 map onto the chapter's §5.2, §5.4 and §5.5. The data overview was a rehearsal for point 3.

Part I: Introduction to NLP

What is Natural Language Processing?

The use of computational techniques to analyse natural language, written and spoken, at scale.

- **Computational:** text turned into digital (numeric) representations, to assess relationships between words, often statistically
- **Natural:** language as humans express it — speeches, social media posts, literature, song lyrics
- **Scale:** beyond a single text, to a million books

Related terms you may have heard

computational linguistics ·
text mining · language
technology · digital
humanities · computational
humanities · distant reading
· quantitative text analysis ·
large language models

The scale problem

The challenge: mass digitisation of analogue resources held in libraries and archives, plus ever-growing amounts of born-digital text — web, social media, user-generated content.

The result: large textual collections, too big to be read by humans.

The opportunity: huge potential in exploring those collections.



Why NLP matters for cultural analytics

Traditional humanities

- Close reading of individual texts
- Deep interpretation
- Limited scale

Computational humanities

- Distant reading of large corpora
- Pattern detection
- Massive scale
- New research questions

Example questions

- How has vocabulary changed over time?
- What topics appear in millions of posts?

- How do different authors use language?
- What is the difference in sentiment between authors?
- What words collocate with a concept?

Part II: The NLP Pipeline

A typical NLP pipeline

Pre-processing → Tokenisation → Lemmatisation → Morphological tagging
→ Part-of-speech tagging → Syntactic parsing → Information extraction → Analysis

Each step makes the next possible, and each one throws something away.

Pre-processing

- Language detection
- **Segmentation**
- Compound splitting
- Punctuation removal
- **Lower-casing**
- **Stop-word removal**
- Spelling standardisation

Language detection — multiple languages affect everything downstream, and matter especially for social media data. Prevents mixing incompatible vocabularies and models.

Segmentation — splitting text into units of analysis: sentences, tokens, paragraphs, chapters, acts of a play. **The unit of analysis decision belongs to the researcher.**

Compound splitting — [notebook](#) → [note](#) + [book](#). Rare for English,

important for morphologically rich languages.

Lower-casing, punctuation, stop words

```
1 text = "Alice was beginning to get very tired of sitting by her sister on the bank."  
2 print(text.lower())
```

```
alice was beginning to get very tired of sitting by her sister on the bank.
```

- **Punctuation removal** strips marks that do not carry meaning for your question
- **Lower-casing** avoids treating **The** and **the** as different words
- **Stop-word removal** drops very common function words — “the”, “and”, “of”

Every one of these throws information away. Lower-casing loses the distinction between **US** and **us**; stop-word removal loses pronouns and negation.

Tokenisation

Splitting text into **tokens** — usually words, but the boundaries are a decision.

```
1 import spacy
2
3 nlp = spacy.load("en_core_web_sm")
4 doc = nlp("Alice wasn't expecting the White Rabbit's waistcoat-pocket.")
5 print([t.text for t in doc])
```

```
['Alice', 'was', "n't", 'expecting', 'the', 'White', 'Rabbit', "'s", 'waistcoat', '-', 'pocket', '.']
```

Types and tokens

```
1 tokens = [t.text.lower() for t in doc if t.is_alpha]
2 print("tokens:", len(tokens))
3 print("types :", len(set(tokens)))
```

```
tokens: 8
types : 8
```

- **Tokens** — every occurrence
- **Types** — every distinct word

The ratio between them is one of the oldest measures in the field.

Lemmatisation

```
1 doc = nlp("The rabbits were running and had run before.")
2 for t in doc:
3     if t.is_alpha:
4         print(f" {t.text:10} -> {t.lemma_}")
```

The	-> the
rabbits	-> rabbit
were	-> be
running	-> run
and	-> and
had	-> have
run	-> run
before	-> before

Reduces words to their dictionary form, so `running`, `ran` and `run` count as one thing.

Part-of-speech tagging

```
1 doc = nlp("The Queen of Hearts painted the roses red.")
2 for t in doc:
3     print(f" {t.text:10} {t.pos_:6} {spacy.explain(t.pos_)}")
```

The	DET	determiner
Queen	NOUN	noun
of	ADP	adposition
Hearts	PROPN	proper noun
painted	VERB	verb
the	DET	determiner
roses	NOUN	noun
red	VERB	verb
.	PUNCT	punctuation

Syntactic parsing and information extraction

```
1 doc = nlp("President Wilson addressed Congress in Washington in 1917.")
2 print("entities:", [(e.text, e.label_) for e in doc.ents])
```

```
entities: [('Wilson', 'PERSON'), ('Congress', 'ORG'), ('Washington', 'GPE'), ('1917', 'DATE')]
```

Typical information-extraction tasks: named entity recognition, relation extraction, event detection, collocation.

Interpretation is key

Every stage of the pipeline is a choice, and every choice removes something.

- Lower-casing removes the difference between **US** and **us**
- Stop-word removal removes pronouns, and **negation**
- Lemmatisation removes tense and number
- Tokenisation decides what counts as a word

Hill, “Contextual Analytics”, §5.3 — representation choices determine which distinctions remain visible and which are suppressed.

Part III: spaCy

spaCy

An NLP library that does the whole pipeline in one call.

```
1 import spacy
2 nlp = spacy.load("en_core_web_sm")
3 doc = nlp(text)
```

- `nlp` is the **model** — a trained pipeline for a language
- `doc` is the **annotated text** — every token, with its lemma, part of speech, and more
- The model is already installed for you; if not, see the module README

From doc to data frame

```
1 import pandas as pd
2
3 doc = nlp("Alice was beginning to get very tired of sitting by her sister.")
4 tokens = pd.DataFrame([
5     {"token": t.text, "lemma": t.lemma_, "pos": t.pos_, "is_stop": t.is_stop}
6     for t in doc if t.is_alpha
7 ])
8 tokens.head(8)
```

	token	lemma	pos	is_stop
0	Alice	Alice	PROPN	False
1	was	be	AUX	True
2	beginning	begin	VERB	False
3	to	to	PART	True
4	get	get	VERB	True
5	very	very	ADV	True
6	tired	tired	ADJ	False
7	of	of	ADP	True

Once it is a data frame, everything from Weeks 3 to 8 applies again.

Word frequencies

```
1 from pathlib import Path
2
3 alice = Path("../data/alice.txt").read_text(encoding="utf-8", errors="replace")
4 doc = nlp(alice[:200_000])
5
6 words = pd.Series([t.lemma_.lower() for t in doc if t.is_alpha and not t.is_stop])
7 words.value_counts().head(10)
```

```
say      481
alice    396
think    134
go        129
little    128
know      107
look      106
begin      93
like       92
come       91
Name: count, dtype: int64
```

Part IV: Live Coding

Live coding

1. Load a text, annotate it with spaCy
2. Tokens, lemmas, parts of speech into a data frame
3. Word frequencies, before and after removing stop words
4. Named entities
5. Loop over the corpus — the Week 9 pattern, applied to text

Before next week

1. Complete `w10_worksheet.ipynb` in the workshop
2. Read Hill, “Contextual Analytics”, §5.3
3. **Work on your report** — it is due 21 January

Note

The quiz in next week’s lecture covers **this week’s** material, and is the **last one**.

Debugging with AI

NLP code fails in a particular way: it runs, and gives you a list of things that are subtly wrong — tokens split oddly, lemmas that are not lemmas, entities that are not entities.

Ask <https://ai.create.kcl.ac.uk/chat> to explain **what a pipeline stage does to your text**, then check its answer against your own output. Do not ask it to interpret your results — that is the part you are being assessed on.



Speaker notes